EXP 13: Case Study on Low-Code Limitations

AIM

To study the limitations of low-code development platforms using Zoho Creator and analyze their impact on application development, customization, performance, integration, and scalability in the Student Admission Management System.

PROBLEM STATEMENT

An educational institution wants to develop a Student Admission Management System using Zoho Creator. The application should manage student details, course preferences, admission applications, document verification, eligibility checks, approvals, notifications, and reports. Although Zoho Creator enables rapid development through visual tools, workflows, and Deluge scripting, complex requirements may affect flexibility and scalability.

SYSTEM REQUIREMENTS

• Student registration and admission application submission.

• Course selection and application status management.

• Document verification and eligibility checks.

• Admission approval and rejection processing.

• Email or SMS notifications to applicants.

• Reports and dashboards for course-wise admissions.

• Integration with payment or student information systems, if required.

PROCEDURE

1. Log in to Zoho Creator and open the Student Admission Management System.

2. Create or review forms for student details, course preferences, admission applications, document verification, and admission status using the drag-and-drop interface.

3. Configure or examine workflows and business rules using Deluge scripting.

4. Identify limitations in advanced customization, complex logic, third-party integrations, and UI design.

5. Analyze performance requirements as the number of student admission records and users increases.

6. Examine integration requirements and platform-specific dependencies.

7. Analyze limitations and identify suitable workarounds or alternative technologies.

8. Document observations and conclude the suitability of low-code development for the Student Admission Management System.

LIMITATIONS IDENTIFIED

1. Limited Customization
Predefined UI components and templates may make highly unique interfaces or interactions difficult without workarounds or external development.

2. Complex Business Logic
Eligibility calculations, course-specific rules, seat allocation, merit ranking, and multi-stage approvals may require extensive Deluge scripting or external services.

3. Platform Dependency (Vendor Lock-in)
The application relies on Zoho Creator tools, data models, and Deluge; migration may require rewriting scripts and workflows.

4. Performance and Scalability
Large numbers of student admission records, frequent report queries, and multiple workflows may affect performance and require optimization.

5. Integration Limitations
Connecting external services may require API configuration, authentication, custom functions, or connectors.

6. Limited Control Over Infrastructure
Zoho Creator manages much of the underlying infrastructure, limiting direct control over server configuration and deployment architecture.

7. Security and Compliance Considerations
Permissions, record access, authentication, and sharing must be configured correctly to protect private information.

8. Licensing and Cost
Additional users, higher usage, advanced deployment features, or integrations may affect subscription costs.

PROPOSED SOLUTION

Limitation	Proposed Solution
Limited customization	Use custom pages and supported HTML/CSS or integrations where appropriate.
Complex business logic	Use optimized Deluge scripts or external services for advanced rules.
Vendor lock-in	Document workflows and maintain exportable data backups.
Performance issues	Optimize student admission reports, queries, and workflows.
Integration challenges	Use supported APIs, webhooks, and connectors.
Infrastructure restriction	Use external services when greater control is required.
Security concerns	Apply least-privilege permissions and regularly audit access.
Cost concerns	Review usage, licensing, and resource requirements.

OBSERVATION

S. No.	Observation	Result
1	Forms created using drag-and-drop tools	Faster development.
2	Basic workflows configured	Less manual coding.
3	Complex logic examined	Additional scripting may be required.
4	External integrations examined	API configuration may be necessary.
5	Performance requirements analyzed	Optimization becomes important as usage grows.
6	Security permissions reviewed	Access can be controlled through appropriate settings.
7	Platform dependency evaluated	Migration may require redevelopment.
8	Licensing requirements reviewed	Cost planning is necessary.

RESULT

The limitations of low-code development in Zoho Creator were studied through the Student Admission Management System case study. Challenges related to customization, business logic, integration, performance, scalability, security, cost, and platform dependency were identified, and suitable solutions were proposed.

CONCLUSION

Thus, the case study on low-code limitations in Zoho Creator was completed successfully. Low-code platforms enable rapid development and simplify workflow automation, but may restrict customization, infrastructure control, complex logic, and portability. Developers should evaluate requirements, workload, integrations, security, and long-term costs. For complex student admission requirements, combining Zoho Creator with custom code and external services can provide a practical solution.